



Data Analytics Project: Railway Gauge Trends

Dataset Used: railway_gauges.csv

This document outlines a structured, end-to-end data analysis project based on railway gauge data. The project is divided into several phases, starting with data collection and preprocessing, followed by exploratory data analysis (EDA), and finally, model building and evaluation. The project is titled **Data Analytics Project: Railway Gauge Trends**.

This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)

Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.

Modules Used:

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (🔗 Easy)

Description:

In this scenario, railway gauge data is imported from a CSV file into a Pandas Data Frame. Missing values are identified and handled to ensure data consistency. Finally, relevant columns are converted into numeric data types to enable accurate analysis and visualization in later stages.

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

```
   Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
0  1964-65         3521        2891         464   6876
1  1965-66         3603        2913         470   6986
2  1966-67         3615        2911         481   7007
3  1967-68         3601        2920         506   7027
4  1968-69         3603        2921         508   7032
Index(['Year', 'Broad Gauge', 'Metre Gauge', 'Narrow Gauge', 'Total'], dtype='str')
```

Figure 1.1: Displaying first 5 rows

Conclusion:

In this task, the railway track dataset was successfully prepared for analysis using **Pandas** through a structured data-cleaning workflow:

The dataset was **loaded into a DataFrame**, The **first five rows and column names** were displayed to understand the dataset's structure and verify available features. **Missing values were identified and replaced with 0**, The gauge-related columns (**Broad, Metre, Narrow, and Total**) were converted into **numeric data types**.

Scenario 2: Simple Visualization (🔗 Easy)

Description:

This scenario focuses on visualizing the growth of the railway network over time using a simple line graph. And Year and Total columns are extracted from the dataset for analysis.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing Total tracks over the Years .
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.

Graph:

The graph is a line plot with Year on the X-axis and Total railway track length on the Y-axis. Each data point is marked and connected to show changes over time. The chart clearly illustrates the progression of the railway network, making it easy to observe increases or decreases across years.

- The **Year** and **Total** columns were extracted to focus specifically on network growth analysis.
- A **line graph** was plotted to visualize how the railway track length changed over time.
- The graph included a **clear title and labeled axes**, improving readability and interpretation.

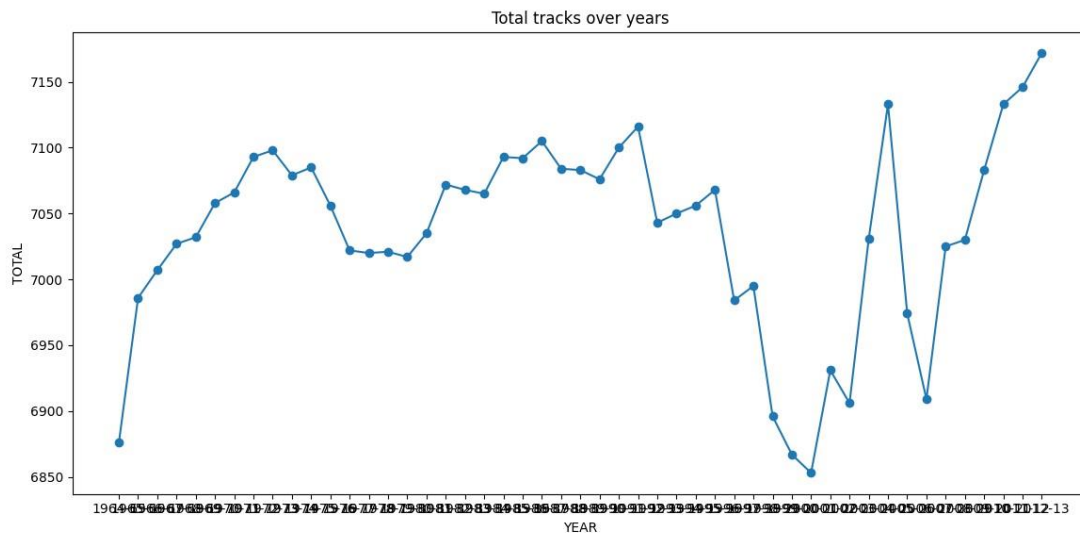


Figure 2.1: Total Tracks Over Years

Conclusion:

The analysis of the **Year** and **Total railway track length** data shows a clear trend in the growth of the railway network over time. After extracting the relevant columns and plotting a line graph, it becomes evident that the total track length has **generally increased across the years**. This indicates steady expansion and development of railway infrastructure, reflecting improvements in connectivity, transportation capacity, and national mobility support.

Scenario 3: Filtering + Bar Chart (🔗 Medium)

Description:

Filter the dataset to get only years after 2000. And select gauge columns from filtered data. After that, plot a grouped bar chart which compares three gauges for selected years. Finally, able to Identify the dominating gauge after 2000.

Task	Description
1.	Filter the dataset to include only years <i>after</i> 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a grouped bar chart comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph:

The graph is a grouped bar chart where each year is represented on the X-axis, and the track length is shown on the Y-axis. For every year, three bars are displayed side by side representing Broad Gauge, Metre Gauge, and Narrow Gauge. The chart makes it easy to compare the contribution of each gauge over time.

- The dataset was filtered to include **years after 2000** for recent trend analysis.
- Only **Broad, Metre, and Narrow Gauge** columns were selected for comparison.
- A **grouped bar chart** was created to visually compare gauge lengths across years.
- Proper **axis labels and legend** were added to improve clarity.

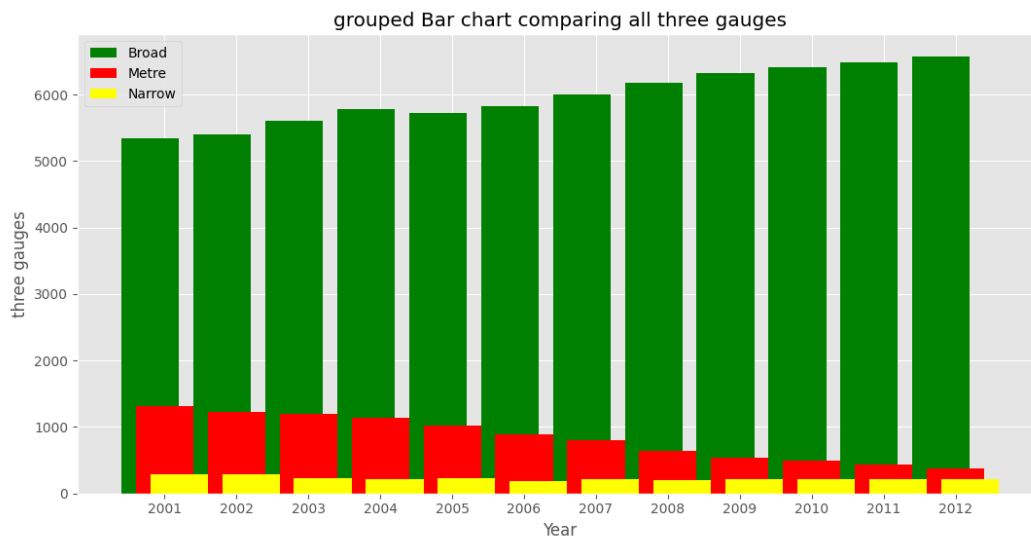


Figure 3.1: Bar Chart

Conclusion:

After filtering the dataset for years after 2000 and comparing Broad, Metre, and Narrow Gauge track lengths using a grouped bar chart, it is clear that **Broad Gauge has become the dominant railway gauge in recent years**. The visualization shows a consistent increase in Broad Gauge track length, while Metre and Narrow Gauge lengths have reduced or remained relatively smaller. This trend reflects modernization efforts and the transition toward a standardized railway system.

Scenario 4: Feature Engineering + Pie Chart (🔗 Medium)

Description:

This scenario focuses on feature engineering to understand the overall contribution of different railway gauges. The total sum of Broad Gauge, Metre Gauge, and Narrow Gauge is calculated across all years in the dataset. These aggregated values are stored in a new structure for easy analysis. A pie chart is then used to represent the proportion of each gauge type. This visualization helps in understanding the historical dominance of each gauge in the railway network.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a pie chart using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Graph:

The graph is a pie chart displaying the percentage contribution of Broad Gauge, Metre Gauge, and Narrow Gauge. Each slice represents one gauge type, with percentage labels shown using `autopct`. The BroadGauge slice is slightly highlighted (exploded) to emphasize its contribution. The chart provides a clear visual comparison of how much each gauge contributes to the total railway network.

- The total length of each gauge type was calculated using **Pandas aggregation**.
- A **Pandas Series** structure was created to store gauge totals.
- A **pie chart** was plotted to show percentage contribution visually.
- Percentage values were displayed using `autopct='% .1f%%'`.

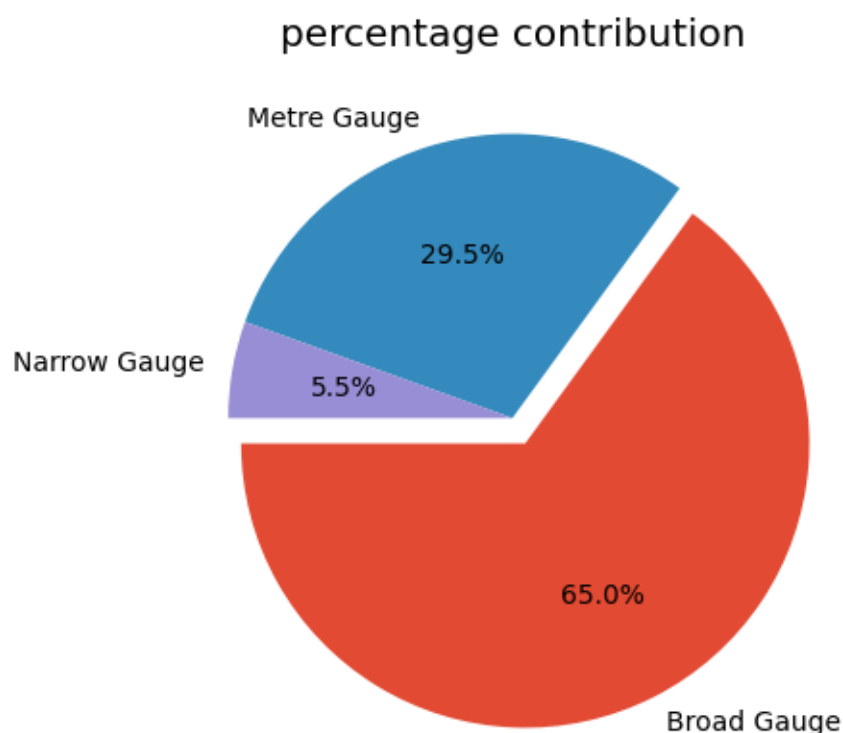


Figure 4.1: Percentage Contribution

Conclusion:

The total lengths of Broad Gauge, Metre Gauge, and Narrow Gauge tracks were calculated across all available years and stored in a Pandas Series. A pie chart was created to visualize the percentage contribution of each gauge type to the overall railway network. The analysis clearly shows that **Broad Gauge contributes the largest share**, indicating its dominant role in the development and expansion of the railway system over time.

Scenario 5: Advanced Analysis + Multiple Graphs (🟡 Hard)

Description:

This scenario focuses on advanced analysis of railway data by combining feature engineering, numerical computation, and multiple visualizations. New percentage-based columns are created to understand the composition of each gauge relative to the total network. NumPy is used to calculate yearly growth, helping identify trends and fluctuations. Multiple graphs, including line plots and stacked bar charts, are used to present both proportional and absolute data. This approach provides a deeper understanding of long-term railway development patterns.

Task	Description
1.	Create new columns for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function <code>np.diff</code> to calculate the yearly growth (change) of the Total tracks column.
3.	Plot: a) A single Line Graph showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A Stacked Bar Chart showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	Highlight/Identify: a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	Final Conclusion: Based on the percentage composition line graph, provide a conclusion to the question: "Is the railway system shifting towards a single dominant gauge?"

Graph:

Two types of graphs are used in this analysis. The first is a line graph showing the percentage composition of Broad, Metre, and Narrow Gauges over time, which helps visualize how each gauge's share changes. The second is a stacked bar chart representing the total composition of all three gauges for each year, showing how they collectively form the entire railway network. Together, these graphs provide both relative and absolute perspectives of the data.

From Percentage Line Graph:

- Broad Gauge percentage increases steadily over time
- Metre Gauge shows continuous decline
- Narrow Gauge becomes minimal in later year

From Stacked Bar Chart:

- Broad Gauge dominates raw track length growth
- Metre Gauge shrinks significantly
- Narrow Gauge nearly phases out in recent years

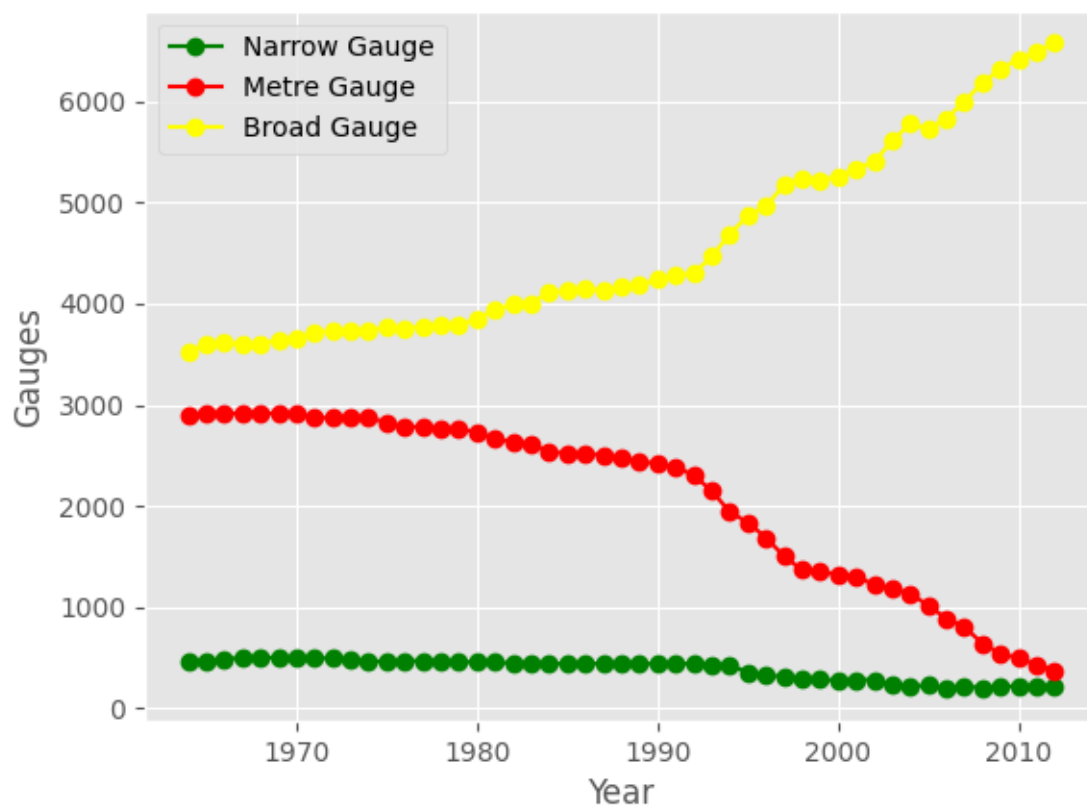


Figure 5.1: Gauges of all Three

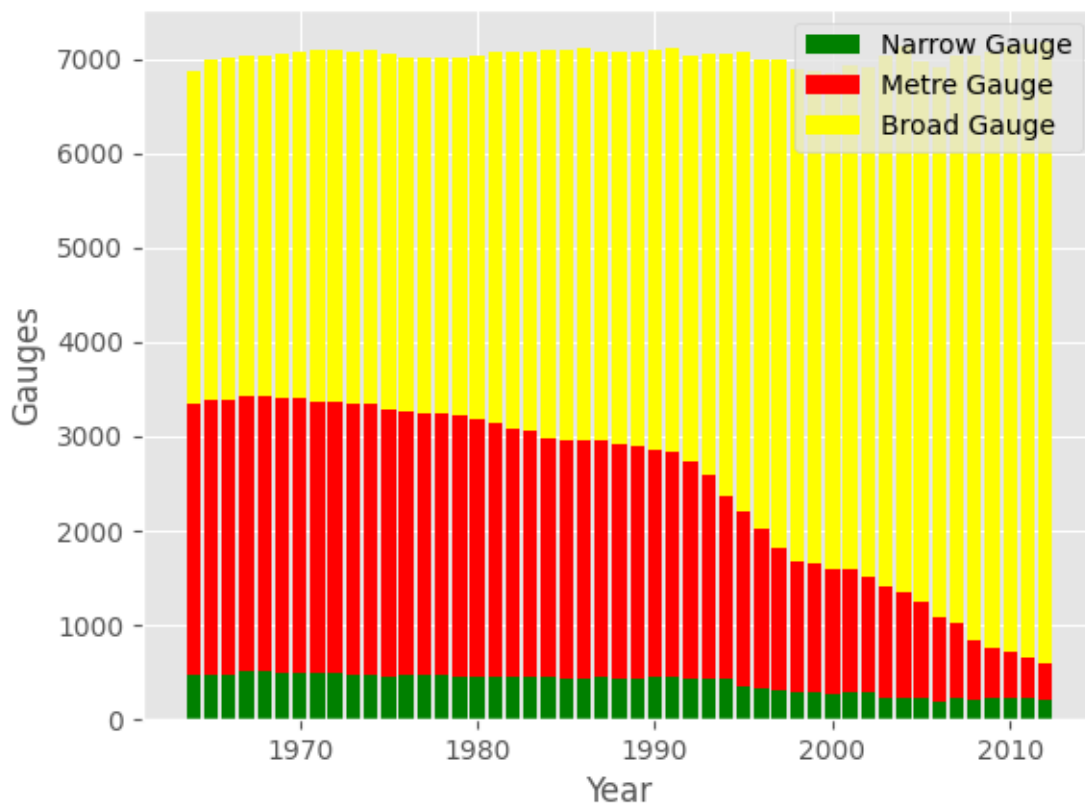


Figure 5.2: Line Graph All Years in Three gauges

Conclusion:

The percentage composition analysis clearly indicates that the railway system is gradually shifting toward a **single dominant gauge, Broad Gauge**. Over the years, the percentage share of Broad Gauge has steadily increased, while Metre Gauge and Narrow Gauge have declined. This trend reflects ongoing gauge conversion projects and modernization efforts aimed at improving efficiency, speed, and interoperability across the railway network. Therefore, the railway infrastructure is progressively transitioning toward standardization around Broad Gauge.

Final Conclusion:

The railway track dataset was successfully analyzed using **Pandas, NumPy, and Matplotlib** to understand the historical growth and composition of the railway network across different gauge types (Broad, Metre, and Narrow). Through systematic data cleaning, transformation, aggregation, and visualization, meaningful trends and structural changes in the railway infrastructure were identified. Percentage composition analysis further confirmed that the railway network is **progressively shifting toward a single standardized gauge system centered on Broad Gauge**. Growth calculations using NumPy also helped identify periods of significant expansion, highlighting phases of major railway development.

What you have learned :

Through this railway network data analysis project, I gained practical experience in **data preprocessing, analysis, and visualization using Pandas, NumPy, and Matplotlib**. The project helped me understand how to transform raw datasets into meaningful insights through structured steps. This project helped me understand how to **clean data, analyze trends, visualize results, and interpret infrastructure growth patterns**, while building strong hands-on skills in **Python data analysis tools (Pandas, NumPy, and Matplotlib)** that are essential for real-world data science and analytics tasks.